

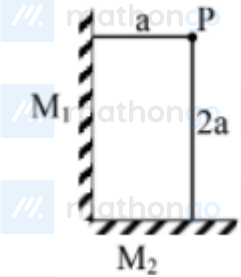
Q1 2021 (31 Aug Shift 2)

Cross-section view of a prism is the equilateral triangle ABC in the figure. The minimum deviation is observed using this prism when the angle of incidence is equal to the prism angle. The time taken by light to travel from P (midpoint of BC) to A is $\dots \times 10^{-10}$ s. (Given, speed of light in vacuum = 3×10^8 m/s and

$$\cos 30^\circ = \frac{\sqrt{3}}{2})$$

Q2 2021 (31 Aug Shift 1)

Two plane mirrors M_1 and M_2 are at right angle to each other shown. A point source 'P' is placed at 'a' and '2a' meter away from M_1 and M_2 respectively. The shortest distance between the images thus formed is : (Take $\sqrt{5} = 2.3$)



- (1) $3a$
- (2) $4.6a$
- (3) $2.3a$
- (4) $2\sqrt{10}a$

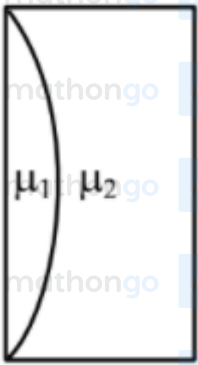
Q3 2021 (31 Aug Shift 1)

An object is placed at the focus of concave lens having focal length f . What is the magnification and distance of the image from the optical centre of the lens?

- (1) $1, \infty$
- (2) Very high, ∞
- (3) $\frac{1}{2}, \frac{f}{2}$
- (4) $\frac{1}{4}, \frac{f}{4}$

Q4 2021 (27 Aug Shift 2)

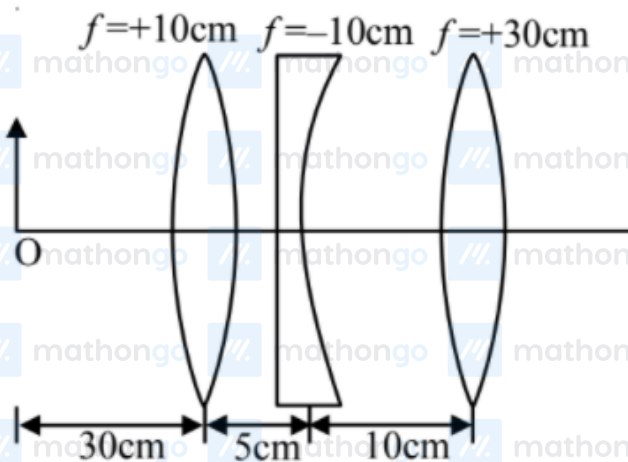
Curved surfaces of a plano-convex lens of refractive index μ_1 and a plano-concave lens of refractive index μ_2 have equal radius of curvature as shown in figure. Find the ratio of radius of curvature to the focal length of the combined lenses.



- (1) $\frac{1}{\mu_2 - \mu_1}$
- (2) $\mu_1 - \mu_2$
- (3) $\frac{1}{\mu_1 - \mu_2}$
- (4) $\mu_2 - \mu_1$

Q5 2021 (27 Aug Shift 1)

Find the distance of the image from object O, formed by the combination of lenses in the figure :



- (1) 75 cm

(2) 10 cm

(3) 20 cm

(4) infinity

Q6 2021 (27 Aug Shift 1)

An object is placed beyond the centre of curvature C of the given concave mirror. If the distance of the object is d_1 from C and the distance of the image formed is d_2 from C , the radius of curvature of this mirror is:

(1) $\frac{2 d_1 d_2}{d_1 - d_2}$

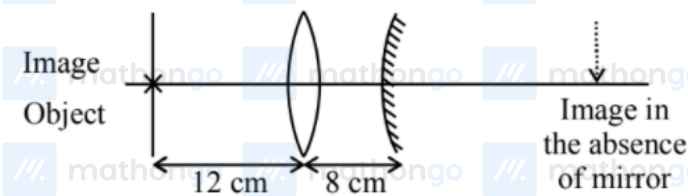
(2) $\frac{2 d_1 d_2}{d_1 + d_2}$

(3) $\frac{d_1 d_2}{d_1 + d_2}$

(4) $\frac{d_1 d_2}{d_1 - d_2}$

Q7 2021 (26 Aug Shift 2)

An object is placed at a distance of 12 cm from a convex lens. A convex mirror of focal length 15 cm is placed on other side of lens at 8 cm as shown in the figure. Image of object coincides with the object.



When the convex mirror is removed, a real and inverted image is formed at a position. The distance of the image from the object will be (cm)

Q8 2021 (26 Aug Shift 1)

Car B overtakes another car A at a relative speed of 40 ms^{-1} . How fast will the image of car B appear to move in the mirror of focal length 10 cm fitted in car A, when the car B is 1.9 m away from the car A?

(1) 4 ms^{-1}

(2) 0.2 ms^{-1}

(4) 0.1 ms^{-1}

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Answer Key

Q1 (5)

Q2 (2)

Q3 (3)

Q4 (2)

Q5 (1)

Q6 (1)

Q7 (50)

Q8 (4)

Q1 (5)

$$i = A = 60^\circ$$

$$\delta_{\min} = 2i - A$$

$$= 2 \times 60^\circ - 60^\circ = 60^\circ$$

$$\mu = \frac{\sin^{-1}\left(\frac{\delta_{\min} + A}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$= \sqrt{3}$$

$$V_{\text{prism}} = \frac{3 \times 10^8}{\sqrt{3}}$$

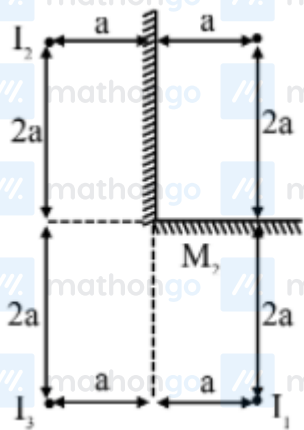
$$AP = 10 \times 10^{-2} \times \frac{\sqrt{3}}{2}$$

$$\text{time} = \frac{5 \times 10^{-2}}{3 \times 10^8} \times \sqrt{3} \times \sqrt{3}$$

$$= 5 \times 10^{-10} \text{ sec}$$

$$\text{Ans} = 5$$

Q2 (2)



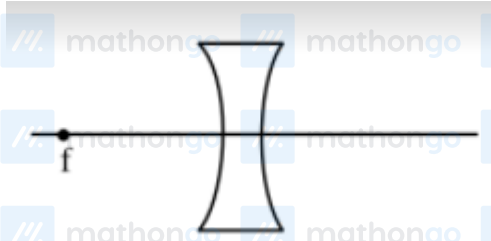
Shortest distance is $2a$ between I_1 & I_3 But answer given is for I_1 & I_2

$$\frac{\sqrt{(4a)^2 + (2a)^2}}{a\sqrt{20}}$$

$$4.47a$$

Option (2)

Q3 (3)



$$U = -f$$

$$\frac{1}{V} - \frac{1}{U} = \frac{1}{-f} \Rightarrow \frac{1}{V} = -\frac{2}{f}$$

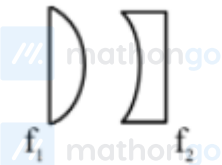
$$V = -\frac{f}{2}$$

$$m = \frac{V}{U} = \frac{1}{2}$$

$$\text{distance} = \frac{f}{2}$$

Option (3)

Q4 (2)



$$\frac{1}{f_1} = (\mu_1 - 1) \left(\frac{1}{R} \right)$$

$$\frac{1}{f_2} = (\mu_2 - 1) \left(-\frac{1}{R} \right)$$

$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{f_{eq}} = \frac{(\mu_1 - 1) - (\mu_2 - 1)}{R}$$

$$\frac{1}{f_{eq}} = \frac{(\mu_1 - \mu_2)}{R}$$

$$\frac{R}{f_{eq}} = (\mu_1 - \mu_2)$$

Q5 (1)

$$\frac{1}{V_1} + \frac{1}{30} = \frac{1}{10}$$

$$\frac{1}{V_1} = \frac{2}{30} \Rightarrow V_1 = 15 \text{ cm}$$

$$\frac{1}{V_2} - \frac{1}{10} = -\frac{1}{10}$$

$$\frac{1}{V_2} = 0 \quad V_2 = \infty$$

$$OV_3 = 75 \text{ cm}$$

Q6 (1)

Using Newton's formula

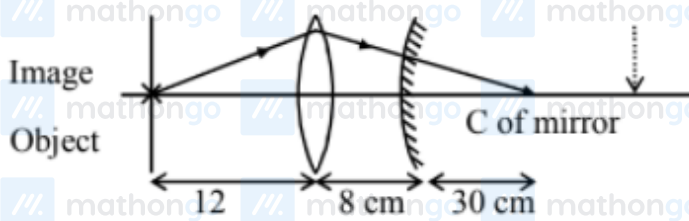
$$(f + d_1)(f - d_2) = f^2$$

$$f^2 + fd_1 - fd_2 - d_1d_2 = f^2$$

$$f = \frac{d_1d_2}{d_1 - d_2}$$

$$\therefore R = \frac{2d_1d_2}{d_1 - d_2}$$

Q7 (50)



For the object to coincide with image, the light must fall perpendicularly to mirror. Which means that the light will have to converge at C of mirror. Without the mirror also, the light would converge at C.

So the distance is : $12 + 8 + 30 = 50 \text{ cm}$

Q8 (4)



Mirror used is convex mirror (rear-view mirror)

$$\therefore V_{I/m} = -m^2 V_{O/m}$$

Given,

$$V_{O/m} = 40 \text{ m/s}$$

$$m = \frac{f}{f-u} = \frac{10}{10+190} = \frac{10}{200}$$

$$\therefore V_{I/m} = -\frac{1}{400} \times 40 = -0.1 \text{ m/s}$$

$\therefore$ Car will appear to move with speed 0.1 m/s.

Hence option (4)